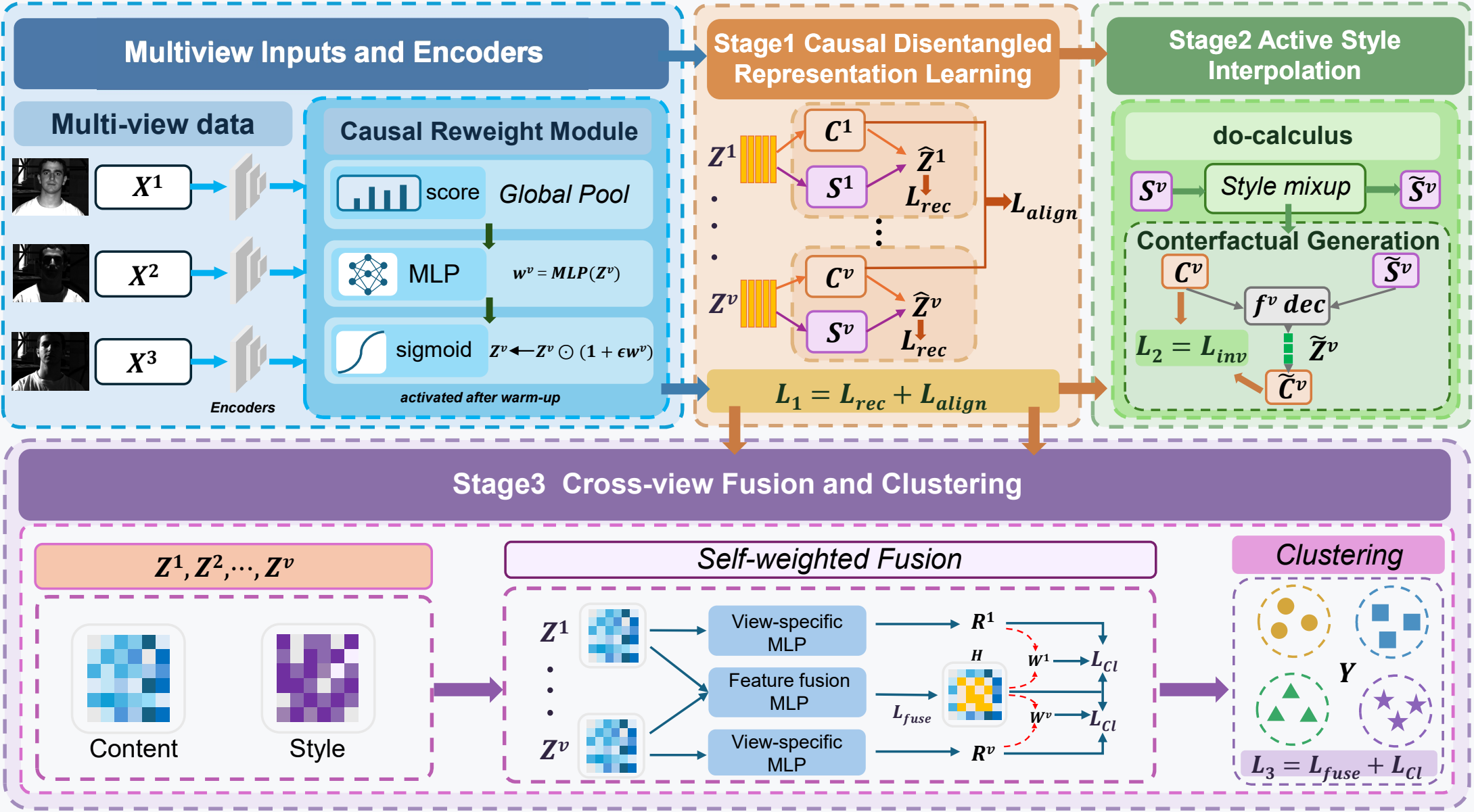
Multiview Inputs and Encoders
Multi-view data
X^1
X^2
X^3
Encoders
Causal Reweight Module
score
Global Pool
MLP
$w^v = MLP(Z^v)$
sigmoid
$Z^v \leftarrow Z^v \odot (1 + \epsilon w^v)$
activated after warm-up
Stage1 Causal Disentangled Representation Learning
Z^1
C^1
S^1
$\hat{Z}^1$
L_{rec}
Z^v
C^v
S^v
$\hat{Z}^v$
L_{rec}
L_{align}
$L_1 = L_{rec} + L_{align}$
Stage2 Active Style Interpolation
do-calculus
S^v
Style mixup
$\tilde{S}^v$
Conterfactual Generation
C^v
$\tilde{S}^v$
$f^v\ dec$
$L_2 = L_{inv}$
$\tilde{Z}^v$
$\tilde{C}^v$
Stage3 Cross-view Fusion and Clustering
$Z^1, Z^2, \cdots, Z^v$
Content
Style
Self-weighted Fusion
Z^1
Z^v
View-specific MLP
Feature fusion MLP
View-specific MLP
R^1
H
R^v
L_{fuse}
W^1
W^v
L_{Cl}
L_{Cl}
Clustering
Y
$L_3 = L_{fuse} + L_{Cl}$